

Hiro Aihara

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EDUCATION

The University of California, Berkeley | B.S. in Data Science

Aug. 2023 – Present

Relevant Coursework: Data Structures & Algorithms, Principles & Techniques of Data Science, Computational Structures in Data Science, Probability for Data Science, Linear Algebra, Calculus, Deep Learning

WORK EXPERIENCE

Credo

May 2026 – Present

Software Engineering Intern

- Implementing Python workflow automation engine using object-oriented design and state machine architecture with 6 node types, helping teams coordinate cross-department processes for faster product launches
- Building Django web application with MongoDB and MongoEngine to store workflow definitions, process instances, and audit logs, helping engineers track requests and meet deadlines more efficiently

Whitney Laboratory

Sep. 2025 – Present

Undergraduate Research Assistant

- Built an automated data preprocessing pipeline standardizing pupil measurements for 200+ participants across 1,000+ trials using pandas z-score normalization, reducing inter-subject variability by 67%
- Developed linear regression analysis quantifying pupil-accuracy relationships ($R^2=0.43$, $p<0.001$) across 3,000+ trials, revealing 23% higher task performance during high-attention periods and validating engagement hypotheses

Lamu

Jan. 2026 – May 2026

Data Engineer Intern

- Designed AWS analytics platform (Lambda, RDS PostgreSQL, QuickSight) tracking 15+ KPIs and automating daily reports, helping the team understand user demographics and improve registration rates
- Engineered ETL pipeline processing 500K+ user records from DynamoDB to PostgreSQL, writing SQL queries to track cohort retention and user growth trends that guided app and website improvements

BLCK UNICRN

Sep. 2025 – Dec. 2025

Software Engineer Intern

- Built Python ETL pipelines processing 50,000+ prompt-response pairs to train company chatbot datasets, achieving 95% data quality through automated validation, helping employees access internal information more quickly
- Engineered validation framework with 12+ automated checks reducing dataset errors by 78%, improving chatbot accuracy so internal teams could get reliable company information faster
- Designed MCP configuration architecture using JSON schemas to control chatbot behavior and response patterns across 8 prompt categories, improving response consistency by 45% through structured data governance

Codify

Sep. 2023 – Sep. 2024

Software Developer

- Optimized Intuit homepage performance by identifying and removing 15+ unused script dependencies and refactoring render-blocking JavaScript code, reducing initial page load time from ~10 seconds to ~5 seconds
- Delivered frontend implementations for 3 enterprise client projects including Intuit, writing 2,000+ lines of production HTML/CSS/JavaScript code that passed client acceptance testing

PROJECTS

TrendFlow

Python, PostgreSQL, Apache Airflow, NLP, Streamlit

- Built real-time trend detection system processing 10,000+ social media posts hourly from Reddit, Twitter, and NewsAPI using AWS Lambda and EventBridge, successfully predicting viral trends 24-48 hours before they peaked
- Optimized database performance from 8 seconds to under 1 second through normalized PostgreSQL schema design and strategic indexing, enabling responsive real-time analytics for trend tracking
- Achieved 99.5% uptime through robust error handling, automated failover, and comprehensive system monitoring

GlobalNews Translation

Python, JavaScript, Flask, NLP, Seq2Seq, Pandas

- Developed a multilingual seq2seq translation model for Japanese-Chinese-English, achieving 85% validation accuracy on 100,000+ sentence pairs, supporting real-time translation across three language combinations
- Deployed Flask REST API serving real-time translations with less than 200ms latency and engineered bilingual training dataset using pandas NLP techniques, improving data quality by 40% through systematic filtering

SKILLS

Programming & Tools: Python, SQL, Java, JavaScript, TypeScript, HTML/CSS, Git, CI/CD

Technologies: AWS (Lambda, RDS, S3, EventBridge, CloudWatch), PostgreSQL, MongoDB, DynamoDB, Django, Flask, Apache Airflow, NLP, Pandas, NumPy, REST APIs, QuickSight

Languages: English, Chinese, Japanese